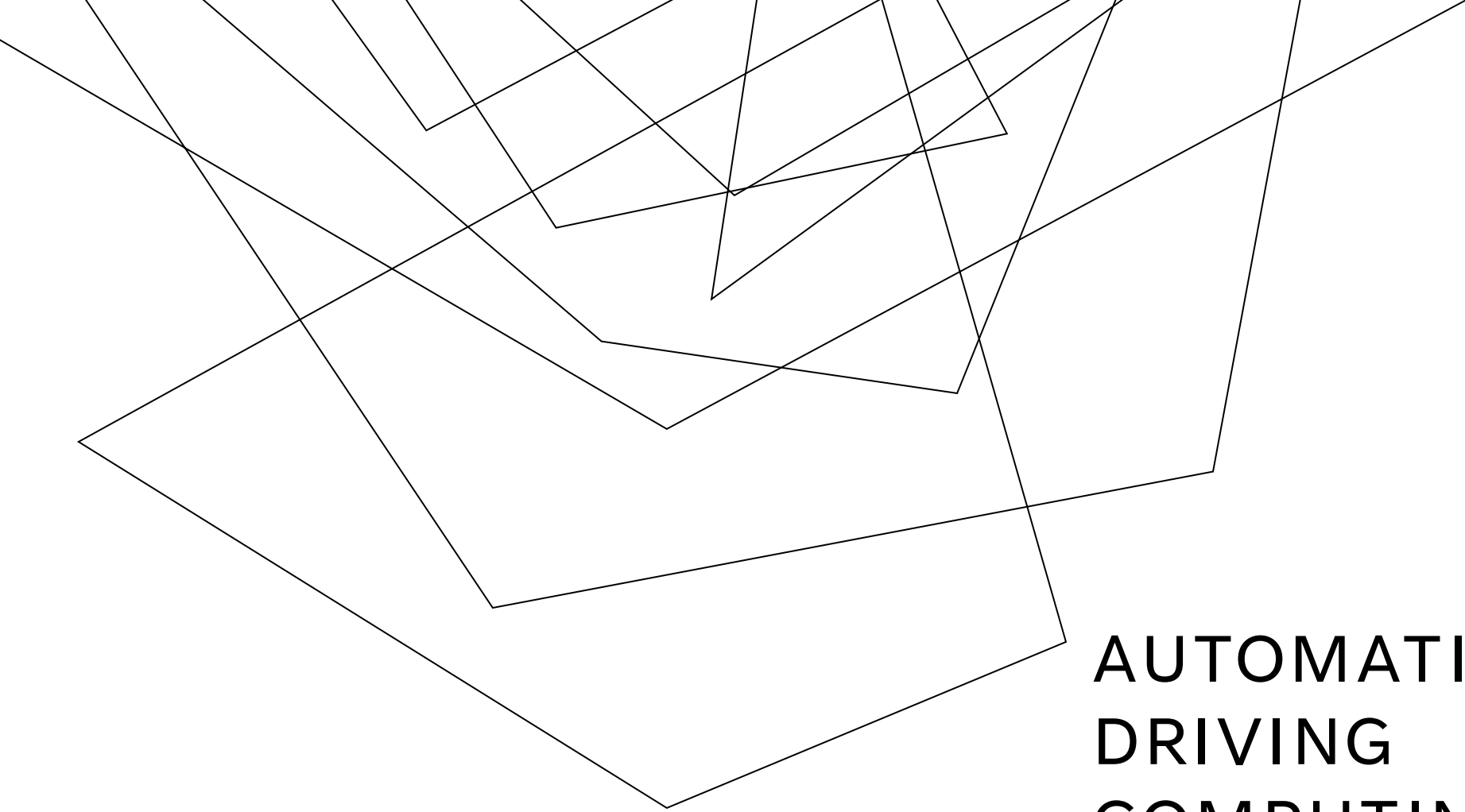


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

10. PERCEPTION (8)





공지 사항

ASSIGNMENT #1 IS OUT

- Due : 10/7 (수) 23 : 59
- 제출 기한이 지난 후에는 추가 제출 (딜레이)은 받지 않음





지난 시간 복습

3D OBJECT DETECTION (2)

1. LiDAR Data Type & Backbone
2. LiDAR-based 3D Object Detection





목차

목차

1. Multi-modal 3D Object detection
2. Perception dataset



An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is steeper, running from the top-center towards the bottom-right. The intersection point is located to the left of the text.

MULTI-MODAL 3D OBJECT DETECTION

MULTI-MODAL 3D OBJECT DETECTION

- Why multi-modal ?
- Image : High-resolution texture information 제공, **but depth 정보 부재**
- LiDAR : Depth 정보 제공, **but low-resolution**
- 각 센서의 장점을 모두 활용할 수 있으면 object detection에 유리
- 각 센서 정보를 어떻게 적절히 fusion할 것인가는 어려운 문제



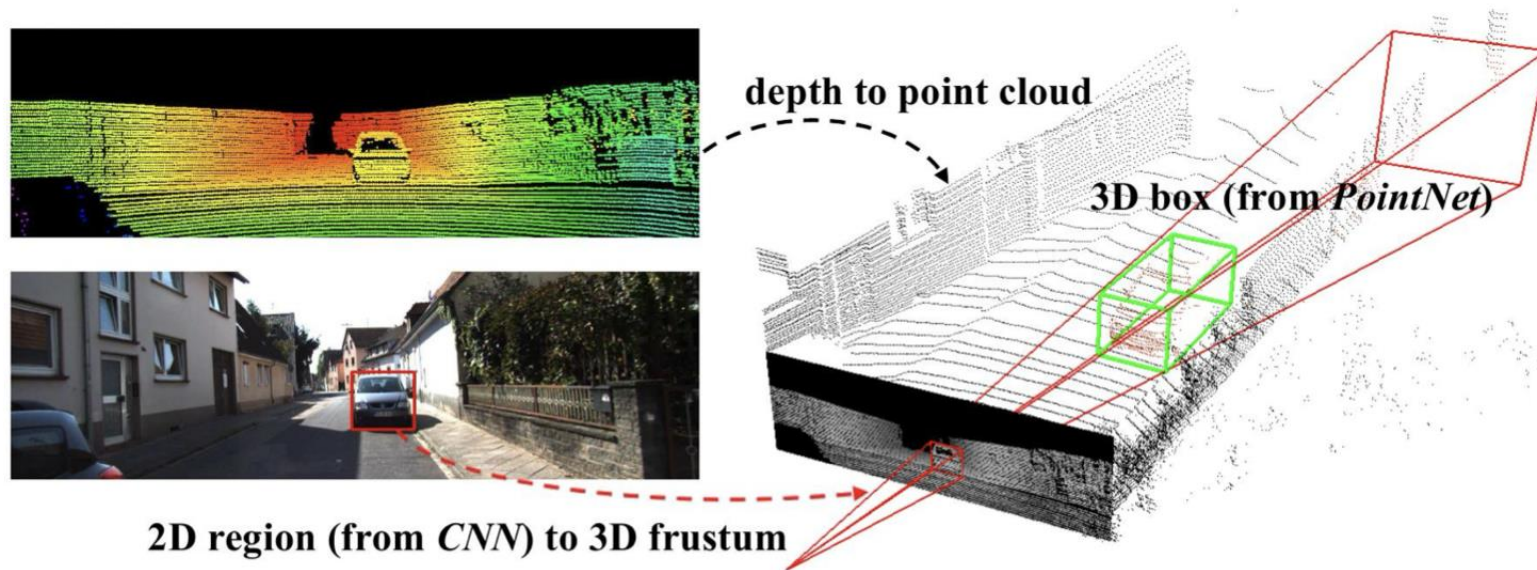
MULTI-MODAL 3D OBJECT DETECTION

- **Frustum PointNets for 3D Object Detection From RGB-D Data (CVPR 2018)**
- **Multi-view 3D Object Detection Network for Autonomous Driving (CVPR 2017)**
 - MV3D



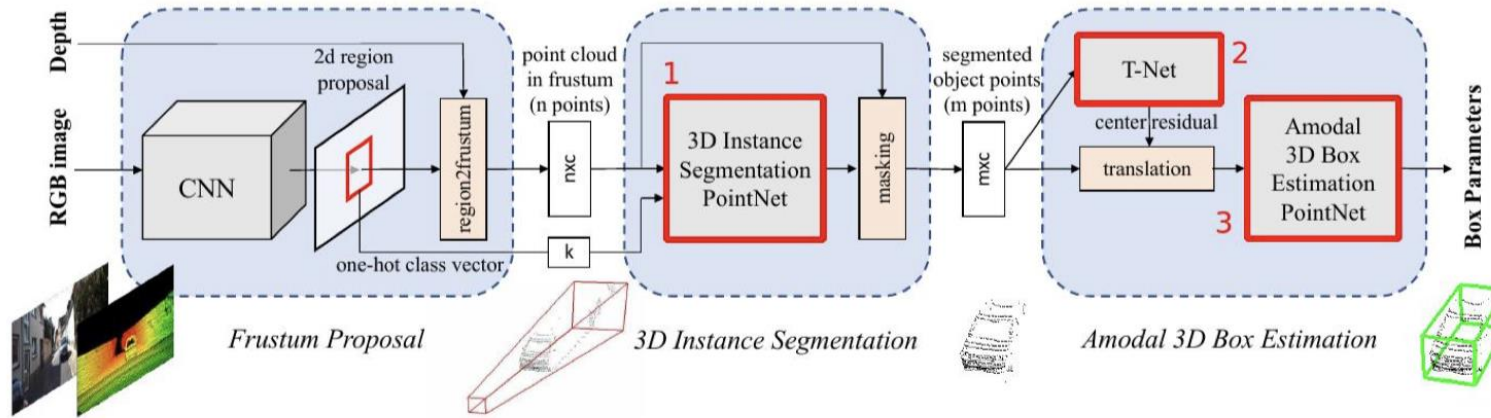
FRUSTUM POINTNET

PointNet의 후속 연구 (PointNet, PointNet++과 동일 저자)



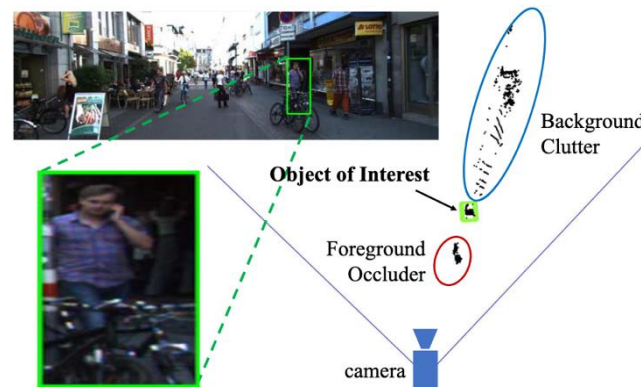
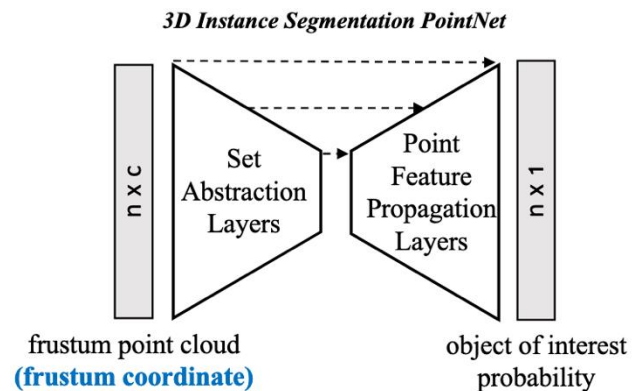
1. Generate **2D object proposals** in the RGB image
2. Extract point cloud from **3D object frustum**
3. **Predict 3D bounding box** from points in frustum via PointNet

FRUSTUM POINTNET

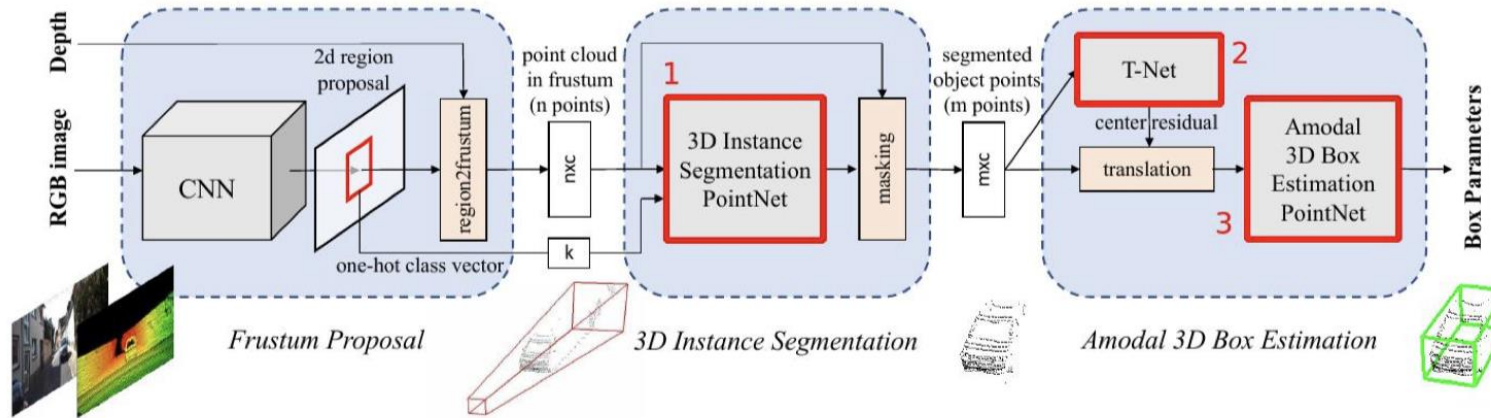


3D box prediction problem is split into **3 stages**: (all implemented via PointNets)

1. **Segmentation** of RGB point cloud (object vs. background)

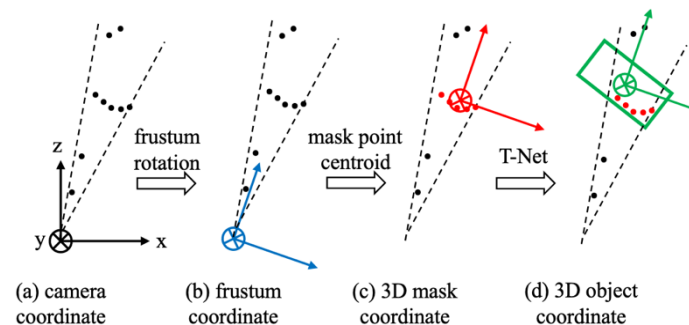
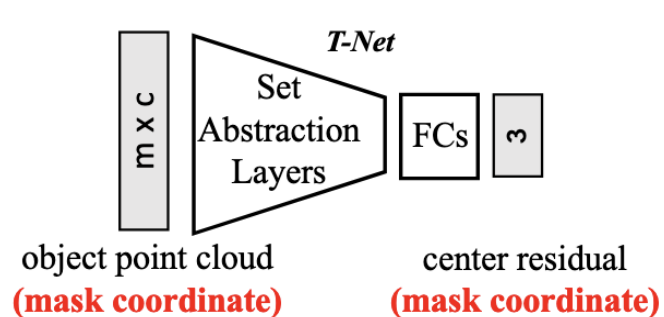


FRUSTUM POINTNET

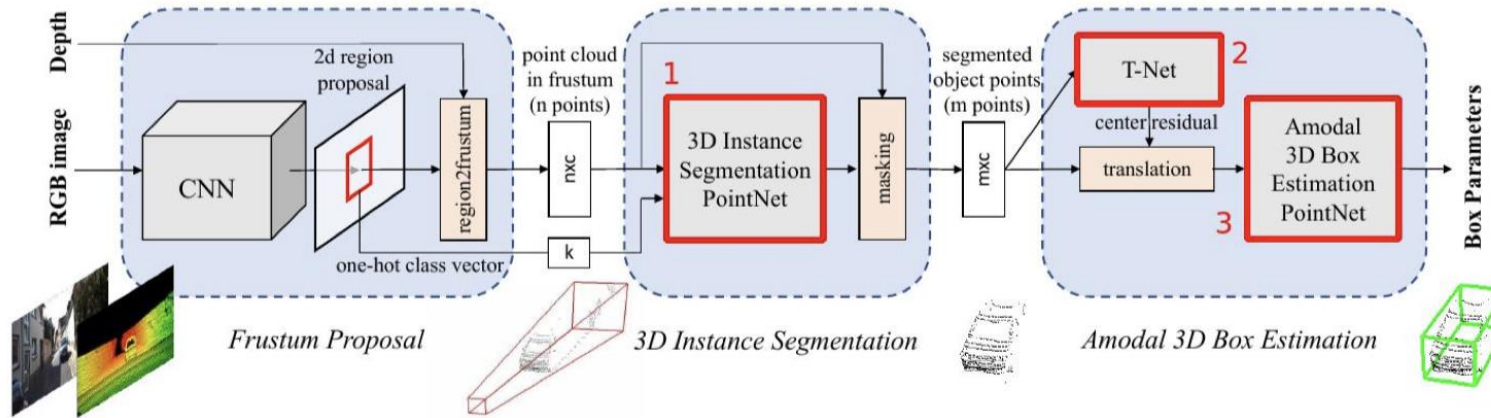


3D box prediction problem is split into **3 stages**: (all implemented via PointNets)

1. **Segmentation** of RGB point cloud (object vs. background)
2. **Translation** of points such that centroid aligns with 3D box center

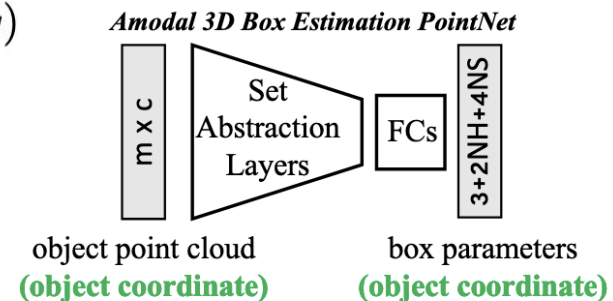


FRUSTUM POINTNET



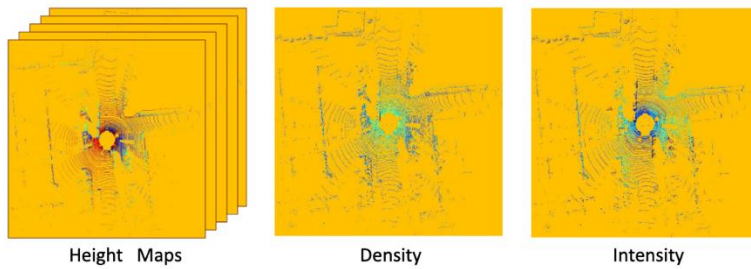
3D box prediction problem is split into **3 stages**: (all implemented via PointNets)

1. **Segmentation** of RGB point cloud (object vs. background)
2. **Translation** of points such that centroid aligns with 3D box center
3. **3D bounding box regression** (x, y, z, w, h, l, y)

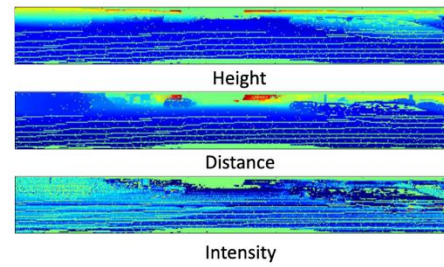


MV3D

- Goal : Multiple 2D view input을 combine & fusion
 - RGB image
 - LiDAR Bird's Eye View (BEV) Projection
 - LiDAR Front View (FV) Projection



(a) Bird's eye view features

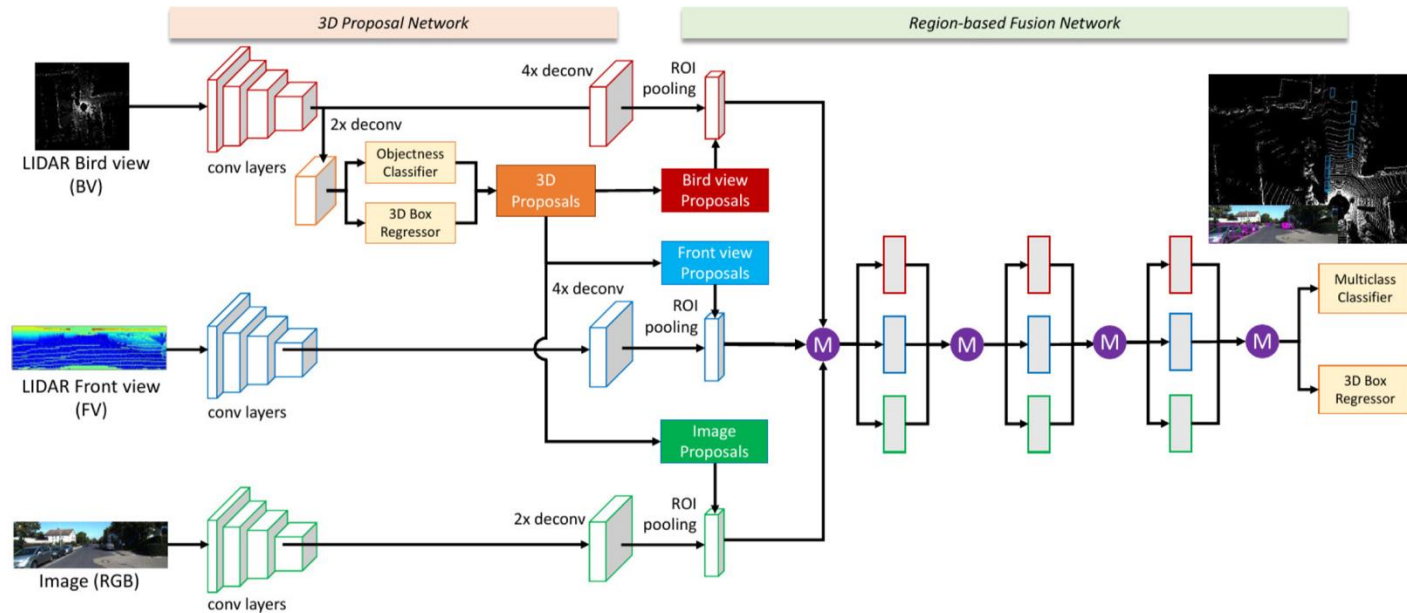


(b) Front view features



(c) RGB image

MV3D



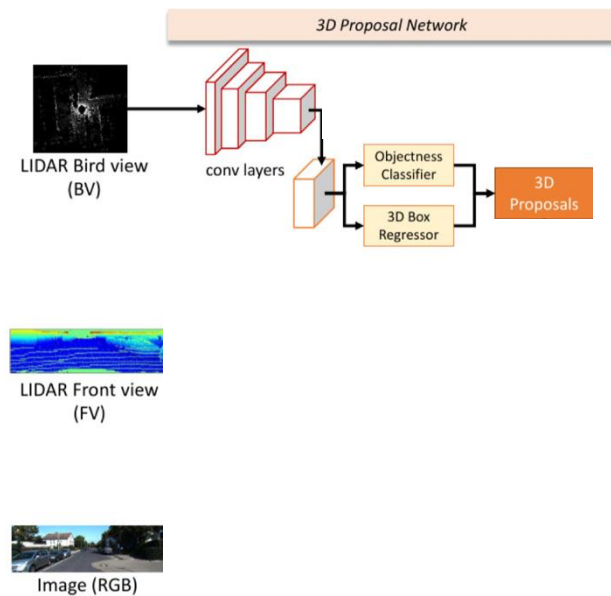
1. Generate 3D proposals from Lidar BV and project into **3 modalities** (Proposals)
2. **Fuse ROI-pooled features** from all modalities, regress box residuals (Refinement)

MV3D

1) LiDAR BEV projection 으로부터 3D Proposal 생성

Why?

- BEV Projection 전후로 object의 physical size가 유지됨
- Object 3D bounding box들이 서로 겹치지 않음

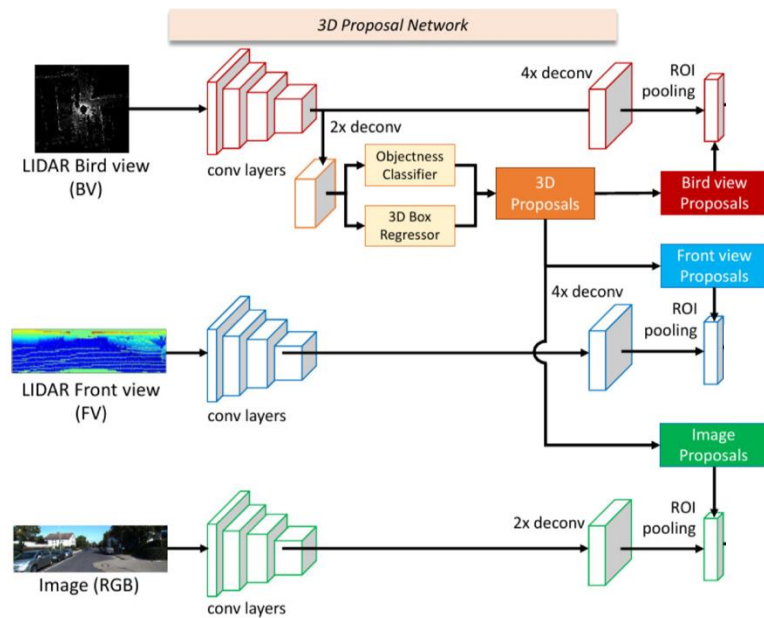


MV3D

2) 생성된 3D Proposal 정보를 각 2D view input 으로 projection 후 ROI pooling

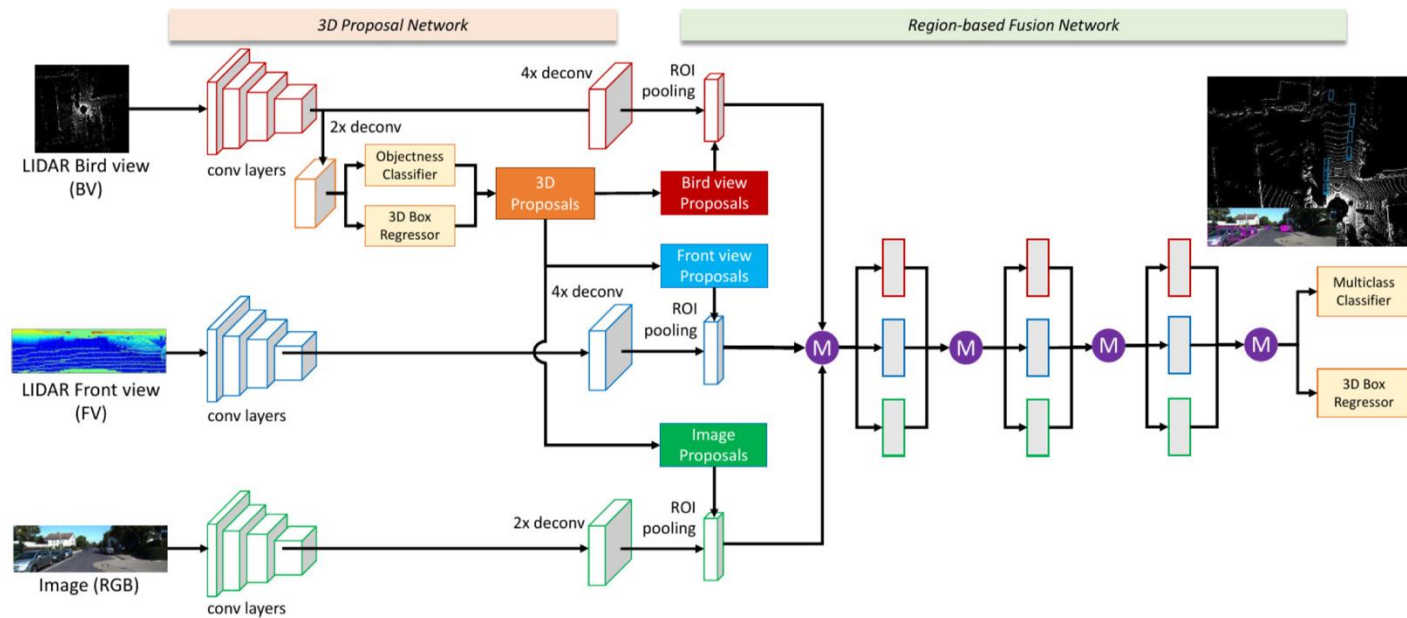
→ 동일한 길이의 feature vector 3개 계산

Cf) 각 modality 별로 resolution 달라서 ROI pooling 필요



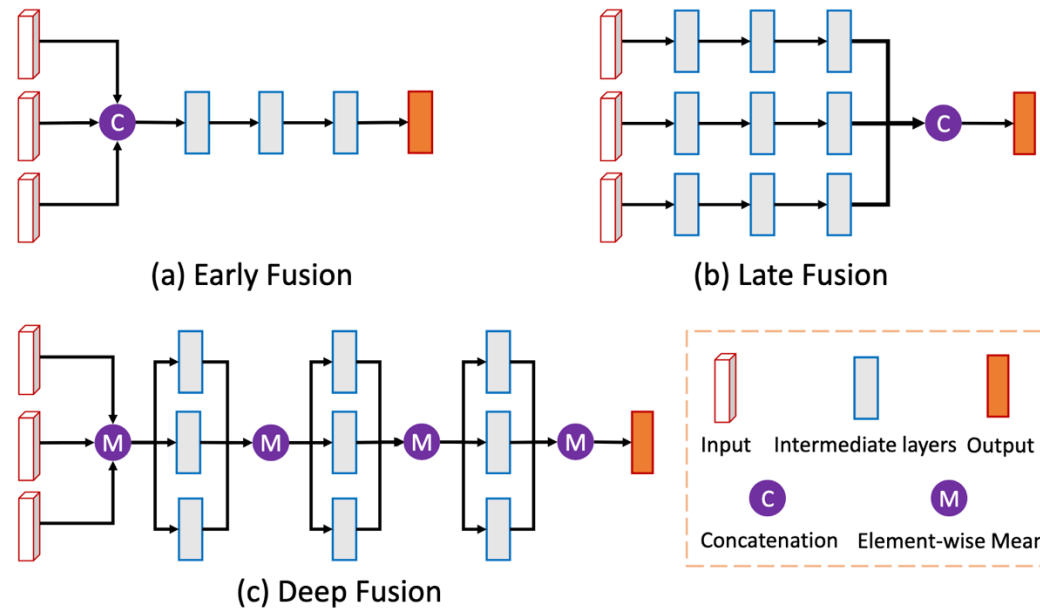
MV3D

3) 각 view input 별로 구한 feature vector를 fusion한 후 최종 classification + regression 수행



MV3D

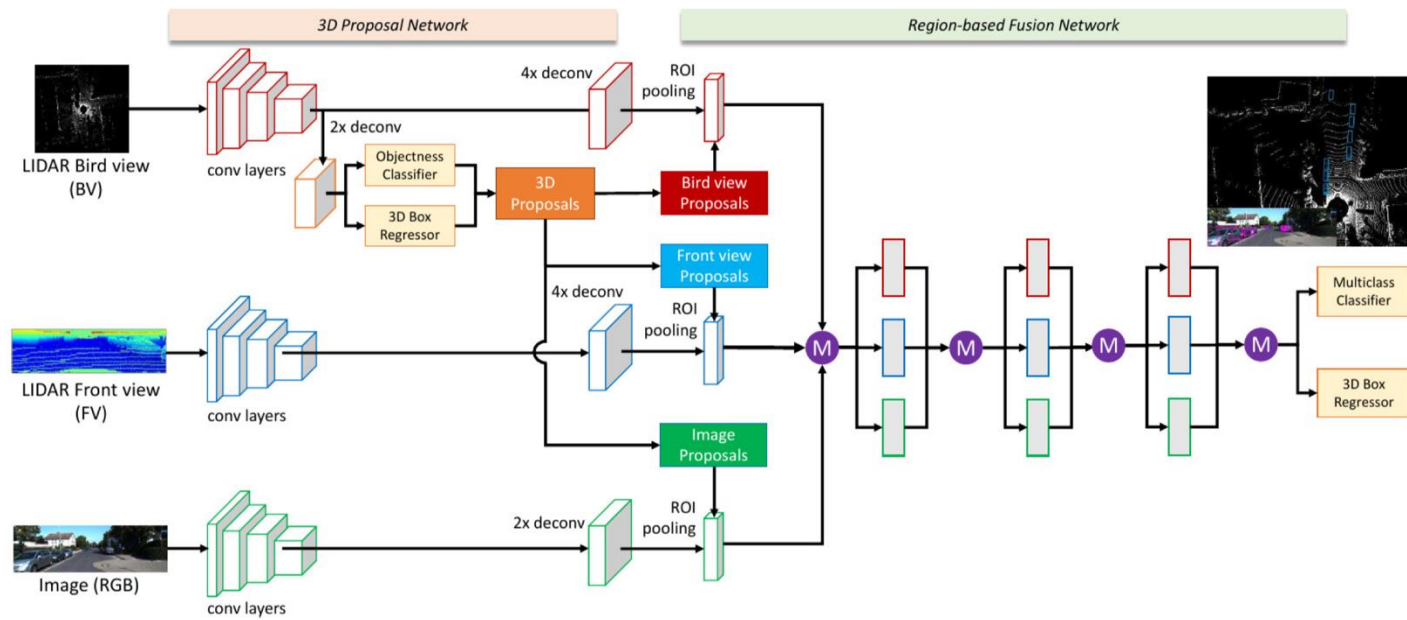
- Feature fusion의 방법
 - Early fusion
 - Late fusion
 - Deep fusion

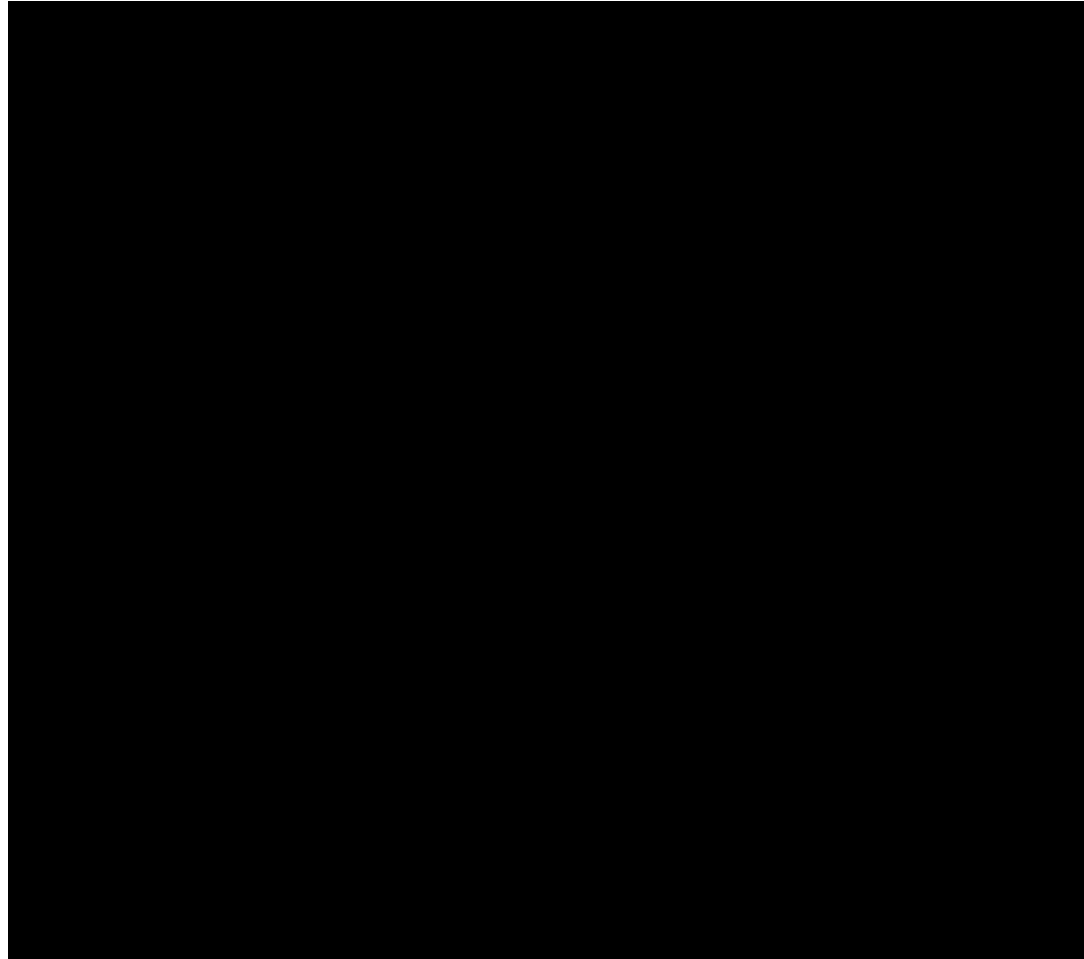


MV3D

전체 pipeline

Two-stage detection pipeline

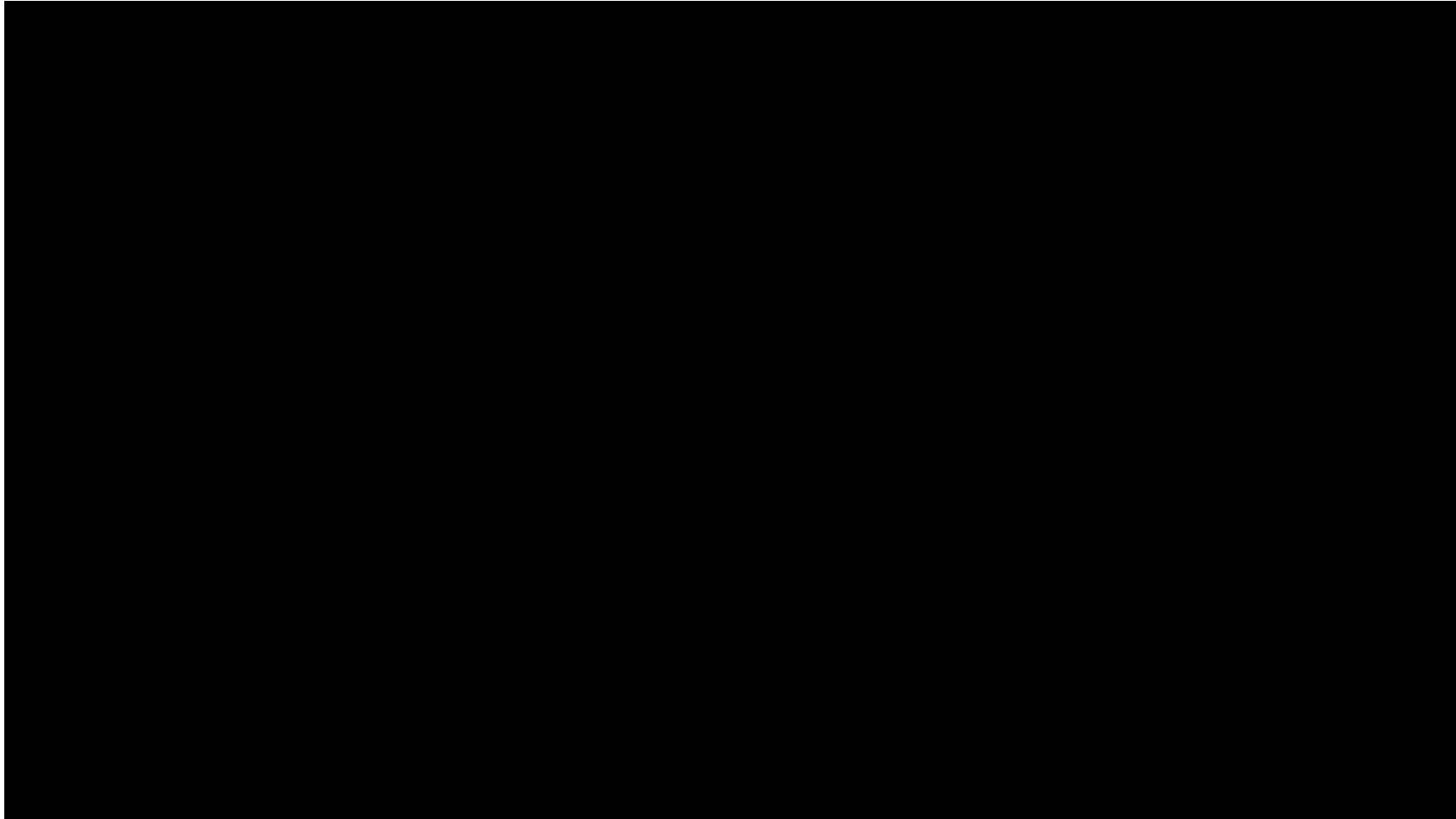




An abstract graphic featuring two thin, dark grey lines that intersect on a light grey background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is more horizontal, sloping slightly downwards from left to right. To the right of the intersection point, the words "PERCEPTION" and "DATASET" are stacked vertically in a bold, black, sans-serif typeface.

PERCEPTION
DATASET

KITTI DATASET (2012)



KITTI DATASET (2012)

- 현재까지 21,000회 이상 인용

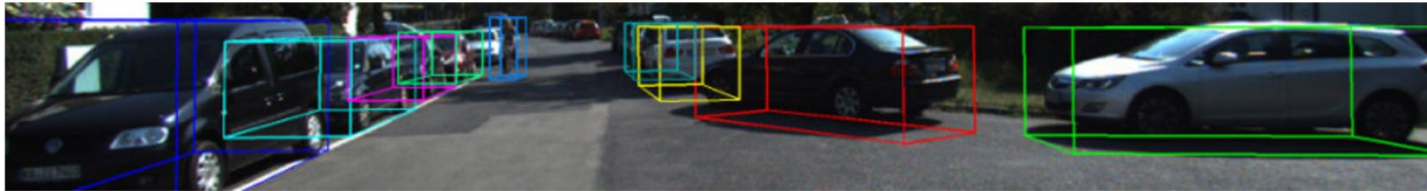
Are we ready for autonomous driving? the kitti vision benchmark suite

[A Geiger, P Lenz, R Urtasun - 2012 IEEE conference on ..., 2012 - ieeexplore.ieee.org](#)

Today, visual recognition systems **are** still rarely employed in ... In this paper, **we** take advantage of our **autonomous** driving ... to 15 cars and 30 pedestrians **are** visible per image). Results ...

☆ 저장 577 인용 21920회 인용 관련 학술자료 전체 20개의 버전

- 7,481 training data / 7,518 test data (Image + LiDAR point cloud)



The 3D object detection benchmark consists of 7481 training images and 7518 test images as well as the corresponding point clouds, comprising a total of 80,256 labeled objects. For evaluation, we compute precision-recall curves. To rank the methods we compute average precision. We require that all methods use the same parameter set for all test pairs. Our development kit provides details about the data format as well as MATLAB / C++ utility functions for reading and writing the label files.



KITTI DATASET (2012)

Leaderboard – Still updating

https://www.cvlibs.net/datasets/kitti/eval_object.php?obj_benchmark=3d

Car

	Method	Setting	Code	Moderate	Easy	Hard	Runtime	Environment
1	VirConv-S		code	87.20 %	92.48 %	82.45 %	0.09 s	1 core @ 2.5 Ghz (C/C++)
H. Wu, C. Wen, S. Shi and C. Wang: Virtual Sparse Convolution for Multimodal 3D Object Detection . CVPR 2023.								
2	UDeerPEP		code	86.72 %	91.77 %	82.57 %	0.1 s	1 core @ 2.5 Ghz (C/C++)
Z. Dong, H. Ji, X. Huang, W. Zhang, X. Zhan and J. Chen: PeP: a Point enhanced Painting method for unified point cloud tasks . 2023.								
3	VirConv-T		code	86.25 %	92.54 %	81.24 %	0.09 s	1 core @ 2.5 Ghz (C/C++)
H. Wu, C. Wen, S. Shi and C. Wang: Virtual Sparse Convolution for Multimodal 3D Object Detection . CVPR 2023.								
4	ViKINET-R			86.04 %	91.20 %	81.18 %	0.06 s	1 core @ 2.5 Ghz (C/C++)
Z. Yu, B. Qiu and A. Khong: ViKINET: Towards Efficient 3D Object Detection with Virtual Key Instance Enhanced Network . CVPR 2025.								

Cf) Frustum PointNet (CVPR 2018)

Method	Cars			Pedestrians			Cyclists		
	Easy	Moderate	Hard	Easy	Moderate	Hard	Easy	Moderate	Hard
DoBEM [35]	7.42	6.95	13.45	-	-	-	-	-	-
MV3D [5]	71.09	62.35	55.12	-	-	-	-	-	-
Ours (v1)	80.62	64.70	56.07	50.88	41.55	38.04	69.36	53.50	52.88
Ours (v2)	81.20	70.39	62.19	51.21	44.89	40.23	71.96	56.77	50.39



ETC

Dataset	Year	Scenes	Size (hr)	RGB imgs	PCs lidar ^{††}	PCs radar	Ann. frames	3D boxes	Night / Rain	Map layers	Classes	Locations
CamVid [8]	2008	4	0.4	18k	0	0	700	0	No/No	0	32	Cambridge
Cityscapes [19]	2016	n/a	-	25k	0	0	25k	0	No/No	0	30	50 cities
Vistas [33]	2017	n/a	-	25k	0	0	25k	0	Yes/Yes	0	152	Global
BDD100K [85]	2017	100k	1k	100M	0	0	100k	0	Yes/Yes	0	10	NY, SF
ApolloScape [41]	2018	-	100	144k	0**	0	144k	70k	Yes/No	0	8-35	4x China
D ² -City [11]	2019	1k [†]	-	700k [†]	0	0	700k [†]	0	No/Yes	0	12	5x China
KITTI [32]	2012	22	1.5	15k	15k	0	15k	200k	No/No	0	8	Karlsruhe
AS lidar [54]	2018	-	2	0	20k	0	20k	475k	-/-	0	6	China
KAIST [17]	2018	-	-	8.9k	8.9k	0	8.9k	0	Yes/No	0	3	Seoul
H3D [61]	2019	160	0.77	83k	27k	0	27k	1.1M	No/No	0	8	SF
nuScenes	2019	1k	5.5	1.4M	400k	1.3M	40k	1.4M	Yes/Yes	11	23	Boston, SG
Argoverse [10]	2019	113 [†]	0.6 [†]	490k [†]	44k	0	22k [†]	993k [†]	Yes/Yes	2	15	Miami, PT
Lyft L5 [45]	2019	366	2.5	323k	46k	0	46k	1.3M	No/No	7	9	Palo Alto
Waymo Open [76]	2019	1k	5.5	1M	200k	0	200k[‡]	12M[‡]	Yes/Yes	0	4	3x USA
A*3D [62]	2019	n/a	55	39k	39k	0	39k	230k	Yes/Yes	0	7	SG
A2D2 [34]	2019	n/a	-	-	-	0	12k	-	-/-	0	14	3x Germany



An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the middle right. The intersection point is located to the left of the word 'SUMMARY'.

SUMMARY

SUMMARY

1. Multi-modal 3D Object detection

- Frustum PointNet
- MV3D

2. Perception dataset



Two thin, dark gray lines intersect diagonally on a light gray background. One line runs from the top-left towards the bottom-right, and the other runs from the top-right towards the bottom-left. They cross each other in the upper-left quadrant of the image.

QUESTIONS?